Madhav Suri

(214) 208-3176 | Surimadhav06@gmail.com | linkedin.com/in/madhav-suri | github.com/madhav-suri | U.S. Citizen

Education

The University of Texas at Dallas Richardson, TX
Bachelor of Science in Computer Science and Mathematics | GPA: 3.8/4.0 *Expected Dec. 2027*

– **Relevant Coursework:** DSA, OOP, Discrete Mathematics, Database Systems, Software Engineering, Operating Systems
– **Certifications:** AWS Cloud Practitioner, Google Project Management Certificate, CAPM in Progress
Portfolio: madhavsuri.com

Skills

Languages: Python, Java, JavaScript, TypeScript, C++, Dart, SQL, Bash, Go, HTML, JSON
Frameworks/Tools: React, Next.js, Node.js, FastAPI, REST APIs, Git, Firebase, Postgres, Docker, Kubernetes, Jenkins, Linux, AWS

Experience

Frontend Developer Jan. 2026 – Present
ACM UTD *Richardson, TX*

– Built and shipped the **React/TypeScript** frontend of an AI-powered academic-support platform used by **350+ live users** across **5+ product workflows**, including scheduling, forum Q&A, self-check grading, and professor analytics.
– Integrated the **Anthropic Claude API** for automated query triage and review workflows, achieving 92% resolution of common support requests without manual routing.
– Developed **45+ reusable UI components** and API-connected dashboard flows while validating backend **REST APIs** and collaborating with ACM engineering teams and academic stakeholders.

Software & AI Engineering Intern Aug. 2025 – Dec. 2025
SoftBank Robotics America *Southlake, TX*

– Developed **EdgeSight**, a real-time assistive vision system running fully on a resource-constrained **NVIDIA Jetson Orin Nano** with 8GB unified memory, Linux/JetPack 6.2, and CUDA 12.6.
– Integrated **YOLOv8** OIV7 600-class detection, multi-object tracking, a multimodal VLM assistant, and a **FastAPI** dashboard, sustaining **15+ FPS** under CPU/GPU memory, power, and thermal limits.
– Reduced on-device VLM query latency by **∼8x** from 74s to **∼7–9s** after benchmarking 5 VLMs, including Gemma-4B, Qwen2.5-VL-3B, moondream, and Gemma-4 E4B/E2B, across **Ollama**, **vLLM**, and **llama.cpp**.
– Deployed a 4-bit-quantized **Gemma-4 E2B** model on **llama.cpp** with FlashAttention and q8_0 KV-cache, while accelerating detection through **TensorRT**, FP16 mixed precision, and MAXN_SUPER clock tuning.
– Profiled GPU-memory contention with **tegrastats**, identified vLLM's ∼2.7GB runtime overhead as infeasible for the 8GB edge device, and designed a hybrid **YOLO + on-demand VLM + dynamic question-routing** pipeline.

Data Engineering Intern May 2025 – Aug. 2025
Riddell *Irving, TX*

– Engineered automated ETL pipelines in **Python** using pandas, NumPy, and SQL to process **5,000+** helmet-impact sensor readings per session into **AWS S3**, automating a previously manual data-preparation process.
– Built **scikit-learn**-powered telemetry reconciliation workflows against performance baselines, improving data quality checks and accelerating product-iteration reporting cycles.

Projects

RiskOS AI | *FastAPI, React, SQLAlchemy, Spark, DuckDB, scikit-learn, Pytest, S3* LINK

– Architected a **fintech fraud-operations platform** using **FastAPI**, **React**, and **SQLAlchemy ORM** to process and route a **6.3M-row PaySim-simulated transaction dataset** into analyst review queues.
– Built an **analytics layer** using **Parquet** datasets in **S3**, queried through **DuckDB** and **Spark**, to support experiment tracking and case-review workflows.
– Implemented **explainable risk scoring** with **scikit-learn Logistic Regression** and 7 deterministic validation rules, achieving **0.901 ROC-AUC**, 82.6% fraud recall, and 94.3% automated routing coverage.
– Added **RBAC**, manager-only overrides, **immutable audit trails**, and a **111-test Pytest** suite in a **GitHub Actions CI/CD** pipeline.

RepoPulse Agent | *Next.js, TypeScript, OpenAI API, GitHub API, Vitest, Docker* LINK

– Engineered an **agentic AI codebase-intelligence platform** in **Next.js** and **TypeScript** with a **7-stage workflow**, audit trail, and human validation gates.
– Built **semantic code search** using **OpenAI embeddings**, syntax-tree parsing, **GitHub API** patch testing, Docker sandboxes, and **200+ Vitest checks**.
– Sustained **sub-1.5s latency** across benchmarked **10,000+ file codebases** while cutting token spend by 35%.

D.I.Y.A. Academic Support Platform | *React, Vite, Node.js, Express, RAG, LangChain* LINK

– Re-engineered an **AI full-stack academic platform** with group-scoped **RBAC** across student and professor routes, securing multi-tenant workflows.
– Built a course-material **Retrieval-Augmented Generation** system using MiniLM embeddings, recursive chunking, cosine search, **Pinecone**, **LangChain**, and OpenAI embeddings across **30+ documents**, achieving 95% retrieval relevance.

Leadership and Activities

Research Team Lead Jan. 2026 – Present
The Erik Jonsson School of Engineering and Computer Science at UT Dallas *Richardson, TX*

– Led undergraduate research under Prof. Yi Ding on **Trusted Execution Environments** and confidential ML inference, building secure C/C++/Python pipelines with **ARM TrustZone** and **Intel SGX**; presented at the Capital of Texas Undergraduate Research Conference; limited overhead to ∼12% across 5 edge-AI workloads and submitted **3 Open Enclave SDK PRs (https://doi.org/10.5281/zenodo.20840404)**.